

1. Explain types of Comments in Python

Comments are used to **explain the code**. Python ignores comments while executing the program.

a) Single-line comment

A single-line comment starts with #.

```
# This is a comment
```

```
name = "Rahul"
```

b) Multi-line comment

Python does not have a special multi-line comment symbol, but we can commonly use **triple quotes** for multiple lines.

```
"""
```

```
This is a
```

```
multi-line comment
```

```
"""
```

```
print("Hello")
```

In Simple Word :

- Single-line comment

""" """ = Commonly used for multi-line comments/documentation

2. What are the 2 ways to execute a Python program?

There are **two common ways**:

a) Interactive Mode

We type and execute Python code **one line at a time**.

Example:

```
>>> print("Hello")
```

```
Hello
```

It is useful for **quick testing and practicing small pieces of code**.

b) Script Mode

We write the complete program in a **.py file** and then run the file.

Example:

```
# hello.py  
print("Hello")
```

Then run:

```
python hello.py
```

In Simple Word :

Interactive Mode - Run **line by line**

Script Mode - Run a **complete .py file**

3. What is a Multi-line Statement?

A **multi-line statement** is a Python statement that is **written across more than one line**.

For example:

```
total = (10 + 20 + 30 + 40)  
print(total)
```

Here, the statement is continued onto multiple lines.

Two common ways:

Using brackets:

```
numbers = [10, 20, 30,  
           40, 50]
```

Using \ (backslash):

```
total = 10 + 20 + 30 + \  
        40 + 50
```

A multi-line statement is a statement that is continued over two or more lines in Python.